Corporate Carbon-VaR: Empirical Mechanics of the Indian Carbon Credit Trading Scheme (CCTS) and EU-CBAM Cross-Border Asymmetries

Kaustav Goswami

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Abstract

This paper establishes a quantitative financial risk framework evaluating the dual regulatory impact of India's Carbon Credit Trading Scheme (CCTS) under the Energy Conservation (Amendment) Act and the European Union's Carbon Border Adjustment Mechanism (EU-CBAM, Regulation EU 2023/956). Using empirical FY2025–26 SEBI Business Responsibility and Sustainability Reporting (BRSR) Principle 6 disclosures across a representative cohort of Indian industrial majors (Steel, Cement, Primary Aluminum), we model corporate balance-sheet exposure across compliance delta thresholds, export tariff penalties, and Marginal Abatement Cost (MACC) arbitrage. We show that domestic regulatory over-compliance fails to protect against cross-border margin erosion, and calculate stochastic Carbon Value-at-Risk (Carbon-VaR) under coupled EU-ETS allowance and Carbon Credit Certificate (CCC) price trajectories.

1 Introduction and Regulatory Architecture

Heavy industrial producers in emerging markets face simultaneous compliance under emerging domestic cap-and-trade regimes and extraterritorial carbon import levies. In India, the Bureau of Energy Efficiency (BEE) operationalized the CCTS to enforce greenhouse gas emission intensity (GEI) targets (tCO_2e per metric ton of output). Concurrently, the European Union's definitive CBAM phase enters financial settlement, requiring authorized declarants to surrender certificates corresponding to direct and indirect embedded emissions.

2 Mathematical and Financial Formulation

2.1 Domestic CCTS Compliance Position

For industrial entity i operating during compliance period t, the net statutory emission position $\Delta E_{i,t}$ (expressed in tCO_2e) is defined by:

$$\Delta E_{i,t} = \left(\text{GEI}_{i,t}^{\text{actual}} - \text{GEI}_{i,t}^{\text{target}} \right) \times Y_{i,t} \tag{1}$$

where $Y_{i,t}$ denotes total verified physical production volume. An entity generates a surplus entitlement of Carbon Credit Certificates ($\text{CCC}_{i,t} = -\Delta E_{i,t}$) if $\Delta E_{i,t} < 0$. If $\Delta E_{i,t} > 0$, the entity enters a deficit liability.

Under Section 26 of the Energy Conservation Act, credit shortfalls unsettled on power exchanges trigger statutory penalty risk:

$$L_i^{\text{CCTS}} = \max(0, \Delta E_{i,t}) \times P_{\text{CCC}} + \max(0, \Delta E_{i,t}^{\text{unsettled}}) \times \left(2 \cdot \bar{P}_{\text{CCC}} \right) \tag{2}$$

2.2 EU-CBAM Tariff Mechanics (Regulation EU 2023/956)

For export volumes directed to EU member states (V_i^{EU}), the gross financial tariff T_i^{CBAM} is determined by embedded direct emissions exceeding EU benchmarks:

$$T_i^{\text{CBAM}} = V_i^{\text{EU}} \times \max\left(0, \text{GEI}_i^{\text{direct}} - \text{BM}_{\text{EU}}\right) \times \left(P_{\text{EU-ETS}} \times \text{FX}_{\text{EUR/INR}} - \delta_{\text{Art9}} \cdot P_{\text{CCC}}\right) \quad (3)$$

where BM_{EU} represents product-specific EU benchmarks (1.370 $\text{tCO}_2\text{e/t}$ for integrated crude steel), and δ_{Art9} represents the recognized Article 9 domestic carbon price deduction coefficient.

2.3 Marginal Abatement Cost (MACC) Arbitrage

A rational corporate treasury allocates between internal abatement capex and exchange purchasing based on net present value:

$$\min_{\mathbf{q}} \sum_{k=1}^{K} c_k q_k + \max\left(0, \Delta E_{i,t} - \sum_{k=1}^{K} q_k\right) P_{\text{CCC}} \quad \text{s.t.} \quad 0 \le q_k \le Q_k^{\max} \quad (4)$$

where c_k is the levelized cost per ton abated for technology k (e.g., WHRS, green hydrogen direct reduction, or captive renewable PPAs).

3 Cohort Simulation Results

Table 1 summarizes empirical SEBI BRSR disclosures and projected EBITDA compression under moderate stress ($P_{\text{CCC}} = 1,200$/t, $P_{\text{EU-ETS}} = 95$/t).

Corporate Entity	Production (Mt)	Direct GEI (tCO_2/t)	CCTS Net (Cr)	CBAM Net (Cr)
Tata Steel Ltd	20.80	2.16	-84.20	+1,142.50
JSW Steel Ltd	23.50	2.24	-112.80	+1,385.20
Jindal Steel & Power	8.40	2.68	+125.40	+684.10
UltraTech Cement	115.00	0.56	-95.00	+42.10
Ambuja Cements	34.00	0.58	-18.50	+0.00
Hindalco Industries	1.35	1.82	+14.20	+188.60

Table 1: SEBI BRSR Principle 6 empirical cohort metrics and modeled annual carbon cost exposure.

4 Conclusion and Risk Implications

Our empirical simulation demonstrates that domestic compliance targets fail to insulate Indian balance sheets from cross-border tax liabilities. Furthermore, exchange illiquidity exacerbates Section 26 statutory penalty cliffs, making internal capital expenditure on low-MACC initiatives mathematically superior to open-market certificate reliance.